Shor’s Algorithm without Qubit Registers — the Photonic Perspective

Johann Pascher

2026

Contents

.1 Shor's Algorithm without Qubit Registers — the Photonic Perspective

Central thesis of this document

Photonically everything is realisable.
Every step of Shor's algorithm — superposition, modular exponentiation, quantum Fourier transformation, measurement — can be physically executed by a photonic system at room temperature.
Not as a simulation, not as an approximation, but as a **direct physical realisation**: the light *is* the superposition, the MZI network *is* the Fourier transformation, the interference *is* the period search.
From the FFGFT perspective this statement is mathematically precisely formulable: the FFGFT field equation on a sub-Planck lattice describes exactly the physical process that delivers the period r as a resonance frequency.

What is not possible: Simulating this field equation on a conventional digital computer — the matrix problem is $N \times N$ in size, hence unsolvable for RSA-relevant N.

What this means: The photonic system solves the problem not through computation, but through physics — exactly as a lens does not compute a Fourier transformation but *is* one.

The quantum computer is often justified by Shor's algorithm for factorisation — the only known algorithm with a genuine exponential quantum advantage over classical methods.

From the FFGFT perspective, the qubit register for Shor is not the most natural way — the natural level is the photonic one.

What Shor Really Does: Only One Quantum Step

Shor's algorithm splits into two fundamentally different parts:

Part 1 — classical: Choose random a, compute gcd(a, N), extract factors from the found period. Ordinary arithmetic — no quantum computing needed.

Part 2 — the only quantum step: Find the **period** r of the function $f(x) = a^x \bmod N$. For this the quantum Fourier transformation (QFT) is employed, which transforms the superposition state:

$$|\psi_3\rangle = \frac{1}{\sqrt{M}} \sum_{j=0}^{M-1} |jr + \ell\rangle \tag{1}$$

into sharp peaks at $y \approx s \cdot 2^n / r$ — from which r is extracted via continued fractions.

The core: period search is a wave operation

The only step in Shor that is exponentially more efficient than classical search is the period search via QFT. Everything else is classical arithmetic.
The QFT itself is a **Fourier transformation**: it finds periodicities through **coherent interference** — constructive amplification at the period frequencies, destructive cancellation of all others.
This is not a discrete computational procedure. This is a **wave operation** — and wave operations are the natural domain of photonic systems, not qubit registers.
An essential difference: The qubit register delivers at each measurement a single probabilistic result — the period appears only through **averaging over many runs**.
A photonic interference system delivers the complete intensity pattern *in one run* directly as a physical quantity — no probabilistic averaging needed.

The QFT is a Wave Operation — Photonic Systems Realise It Physically

A Fourier transformation through coherent interference is exactly what every wave system does by nature.

A **lens** performs an optical Fourier transformation — in the time light takes to pass through it. No register, no gate, no cooling.

A **grating spectrometer** decomposes light into its frequency components through physical interference — this is a physical Fourier analysis that instantly makes periodicities in the light field visible.

An **MZI network** with n stages implements geometrically an n-point Fourier transformation — the phase relations between the paths correspond exactly to the twiddle factors $e^{2\pi ijk/n}$ of the QFT.

FFGFT: QFT as physical interference of torsion waves — mathematical formulation

In FFGFT the QFT is not an abstract unitary matrix on a qubit register — it is the natural dynamics of a wave system.
The mathematical formulation:
The photonic torsion field $\Psi(\mathbf{x}, t)$ in the waveguide satisfies the FFGFT wave equation:

$$\left[\nabla^2 - \frac{1}{c^2}\partial_t^2 + \xi \cdot K(\mathbf{x})\right]\Psi = 0, \tag{2}$$

where $K(\mathbf{x})$ encodes the crystal geometry of the MZI network. The stationary solutions of this equation are exactly the eigenmodes of the QFT — the twiddle factors $e^{2\pi ijk/n}$ appear as natural phase relations between the modes of the waveguide.
What this means: This equation is *not simulable on a conventional digital computer* — the state space problem is exponentially large. But it is *physically exactly solvable*: the photonic system solves it by relaxing into the equilibrium state.
This is naturally given at the ξ level $N \approx 9$ (photonic resonator, eV scale, room temperature) — the torsion waves interfere collectively in the crystal geometry of the waveguide.
A qubit register at ξ level $N \approx 8$ (mK, spin qubits) *emulates* this process digitally through gate operations — a detour via a colder, more fragile level to do something the warmer level does directly.

The MZI Network: Photonic QFT is Principally Fully Realisable

Photonic QFT: principally complete

A photonic realisation of Shor's QFT step is **principally fully possible** — not as a simulation, but as a direct physical evaluation through wave interference.
The MZI network realises the quantum Fourier transformation in the time light takes to traverse the network — at room temperature, without cooling, without qubit register.
The only open problem is the **input preparation**: the modular exponentiation $f(x) = a^x \bmod N$ must be encoded for all x simultaneously in the light signal. Precisely for this the eigenvalue problem (Section .2) is the proposed solution.

The quantum Fourier transformation computes:

$$F(k) = \sum_{x=0}^{N-1} f(x) \cdot W_N^{kx}, \quad W_N = e^{2\pi i/N}. \tag{3}$$

The factors $W_N^{jk} = e^{2\pi ijk/N}$ are called **twiddle factors** — they describe phase rotations by angle $2\pi jk/N$.

In an **MZI network** (Mach-Zehnder interferometer) exactly this phase rotation is physically realised: the light is split by a beam splitter into two paths. Through changing the optical path length (phase shifter, piezo, refractive index modulation) one path experiences the delay $\Delta\phi = 2\pi jk/N$ relative to the other.

The outputs of the MZI correspond exactly to the **butterfly operations** of the classical FFT (Cooley-Tukey algorithm): two input waves are mixed, multiplied by the twiddle factor and distributed to two outputs. An n-stage MZI network cascades these operations — the total network computes a 2^n-point FFT in the time light traverses the network.

FFGFT: MZI network as torsion wave QFT

In FFGFT every phase rotation by $\Delta\phi$ is a rotation of the torsion configuration by the same angle in the transverse plane.
The MZI network does not *emulate* the mathematics of the QFT — it *is* physically a QFT: the phase relations between the paths are the twiddle factors, the beam splitters are the

butterfly operations, and the coherence of the light ensures the superposition.
At the ξ level $N \approx 9$ (photonic, eV scale) this operation is thermally stable at room temperature — in contrast to the digital emulation at level $N \approx 8$ (spin qubit, meV scale, mK cooling mandatory).
The qubit register emulates digitally what the MZI network directly does physically — a detour via a fragile, cold level for an operation that the warmer level naturally masters.

What the MZI Network Does *Not* Solve

The MZI network realises the QFT — but only for a *known, complete* input vector.

The actual bottleneck in Shor is the evaluation of $f(x) = a^x \bmod N$ for all $x = 0 \ldots 2^n - 1$ simultaneously. A 2^n input vector does not exist as a physical light signal — it would first have to be computed, which costs $O(N)$.

The eigenvalue problem (Section .2) is the attempt to circumvent this hurdle too: not computing all $f(x)$, but reading off the period as a resonance frequency of the potential.

FFGFT-Shor: ξ Resonance Replaces the QFT

Doc. 147 develops a FFGFT-Shor algorithm that replaces the QFT through direct **ξ-resonance search**:

$$\text{Resonance}(r) = \frac{\exp(-\xi \cdot r^2 / D_{\text{frak}})}{|a^r \bmod N - 1| + 1}. \tag{4}$$

Strong resonance at an r means $a^r \equiv 1 \pmod{N}$ — the sought r is found.

Why No Qubit Register is Needed — and What Registers Can Still Do

FFGFT: Shor is a wave interference problem, not a discrete state problem

The period r of $a^x \bmod N$ is a property of the *entire function* $f : x \mapsto a^x \bmod N$ — not of an individual value $f(x_0)$.
It is a **global spectral property**: the frequency with which the function returns to the value 1.
A discrete computational machine that evaluates $f(x)$ sequentially needs $O(r)$ steps to find the period — exponential in the problem size.
A wave system that encodes $f(x)$ simultaneously for all x (as amplitude, phase, or intensity) can extract the period through coherent interference in a single pass — regardless of the value of r.
That is exactly what the QFT in the qubit register does — but the qubit register is not the only way. A photonic MZI network does the same, at room temperature, without cooling, without qubit coherence.

Two problems for QC, one for photonics

Qubit register:

- Problem 1: Decoherence in mK environment, error correction required
- Problem 2: Modular exponentiation expensive ($O(n^3)$ gates)
- Advantage: Complete quantum parallelism in one system

Photonic MZI network:

- Problem 1 solved: room temperature, no decoherence
- Problem 2 open: input preparation for $f(x)$ for all x
- Advantage: QFT directly, without gates, in one light pass

Quantum computer is no better calculator

The quantum computer does not compute faster by having faster bits — it computes differently: through quantum parallelism, superposition and interference.
The photonic system is an even more direct realisation: it does not *compute* the interference — it *is* interference. The period r appears as a physical resonance, not as a numerical result.

.2 FFGFT-FFT as Eigenvalue Problem — the Sub-Planck Lattice

From Summation to Resonance Problem

The standard Shor algorithm computes:

$$\text{QFT} : \quad |x\rangle \mapsto \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i x k/N} |k\rangle . \tag{5}$$

The FFGFT reformulation replaces this by an eigenvalue problem: instead of transforming a state vector, the period is sought as the eigenfrequency of a tight-binding Hamiltonian:

$$-\frac{\Psi_{n+1} - 2\Psi_n + \Psi_{n-1}}{\ell_0^2} + \frac{E^2}{\hbar^2 c^2}\left(1 - \xi \cdot \frac{a^n \bmod N}{N}\right)\Psi_n = E\Psi_n, \quad n \in \mathbb{Z}. \tag{6}$$

Fundamental change of problem class

The standard QFT transforms: **State vector → frequency space**.
The FFGFT formulation asks: at which energies does the potential $V(n) = -\xi \cdot (a^n \bmod N)/N$ have **localised eigenstates**?
This is not a computational problem — it is a **spectroscopic measurement problem**.

The Sub-Planck Scale Makes the Potential Consistently Discrete

On the sub-Planck lattice with $\ell_0 = \xi \cdot \ell_P \approx 2 \times 10^{-39}$ m the lattice is fine enough to represent every jump of a^n mod N without information loss.

Sub-Planck scale: discreteness as strength

What appears as a problem in the continuum — the jumps of a^x mod N — becomes on the discrete lattice a **coupling matrix of a tight-binding model**:

$$H_{mn} = -\frac{1}{\ell_0^2}\left(\delta_{m,n+1} + \delta_{m,n-1} - 2\delta_{m,n}\right) + \frac{E^2}{\hbar^2 c^2}\left(1 - \xi \cdot \frac{a^n \bmod N}{N}\right)\delta_{m,n}. \quad (7)$$

This is a **tight-binding Hamiltonian** with a quasiperiodic potential — a model class well understood in solid-state physics (Anderson localisation, Hofstadter butterfly).
The "jumpiness" of a^n mod N corresponds to a pseudorandom potential. The question shifts from "Do the eigenvalues exist?" to "Are the eigenstates localised at the right energies?" — a question of **Anderson localisation**.

The Open Conjecture

FFGFT-FFT conjecture (open)

For every $N = p \cdot q$ and every a coprime to N, there exists $\xi > 0$ and a lattice constant $\ell_0 \ll \ell_P$, such that the discrete eigenvalue problem (Eq. 6) has localised eigenstates at $E = 2\pi\hbar c \cdot k/r$ if and only if r is the order of a modulo N.

Status: Unproven. This conjecture is mathematically precisely formulable but not yet proven or refuted. It is not simulable on a classical computer (the matrix problem is $N \times N$ in size), but physically realisable in a FFGFT torsion field on sub-Planck scale.

What the Formulation Achieves — and What Not

Honest assessment

What this formulation achieves:
It gives the FFGFT-FFT idea a mathematically precise form. It connects the number-theoretic problem (order of a modulo N) with a physical resonance problem (localised eigenstates of a tight-binding model). It is basis-system independent and not reliant on floating-point artefacts.

What this formulation does not achieve:
It does not prove that the eigenstates are localised at the right energies. It is not more efficient on a classical computer than brute-force period search — the Hamiltonian problem is just as large as the original. The sub-Planck scale makes the problem more consistent, but not simpler.

The real contribution:
A physical realisation — a FFGFT torsion field that physically solves Eq. (6) — would solve the problem in the time the field takes to find its equilibrium state. This is not an algorithmic optimisation, but a change of realisation substrate. Just as an optical spectrometer performs a Fourier transformation at the speed of light — not through algorithm, but through physics.

.3 Empirical Findings: What the Test Programmes Show

Parallel to the mathematical formulation, four interactive test programmes were developed and run (BigInt-exact, no float artefacts). All programmes are publicly accessible at:

https://github.com/jpascher/T0-Time-Mass-Duality/tree/main/2/html

`t0_shor_bigint.html` BigInt-corrected Shor simulator — exact modular arithmetic for arbitrary N, comparison FFGFT period search vs. classical trial division, benchmark with 10 test cases from $N = 15$ to $N \approx 10^{18}$.

`t0_shor_hypothesis.html` ξ-basis hypothesis test — compares ξ-optimal bases with random bases on period minimality, statistical test over many N with win rate and median ratio r_{random}/r_ξ.

`t0_xi_num_algebraisch.html` $\xi_{\text{num}} = \lambda(N)/N$ distribution analysis — cluster analysis over 200 semiprime N, comparison with FFGFT level hierarchy ξ^k.

`t0_primzahl_periodizitaet.html` Prime gaps, residue class patterns, $p-1$ smoothness and autocorrelation of the gap sequence — four tests on the period structure of the primes themselves.

Finding 1: ξ_{num} is Algebraic, not Geometric

The quantity $\xi_{\text{num}}(N) = \lambda(N)/N$ shows over 200 random semiprime N three to five accumulation points:

ξ_{num}	Share	Cause
$\approx 1/2$	55 %	$\text{gcd}(p-1, q-1) = 2$
$\approx 1/4$	14 %	$\text{gcd}(p-1, q-1) = 4$
$\approx 1/6$	13 %	$\text{gcd}(p-1, q-1) = 6$
$\approx 1/12$	6 %	$\text{gcd}(p-1, q-1) = 12$

This is the harmonic series — not a FFGFT resonance structure, but the **divisor structure of even numbers**. All primes > 2 are odd, so $p-1$ and $q-1$ are always even. The gcd is always a multiple of 2.

ξ_{num} is structurally different from ξ

$\xi = 4/30000$ comes from the 3D tetrahedron geometry of physical space.
$\xi_{\text{num}}(N) = \lambda(N)/N$ comes from the algebraic divisor structure of the prime factors of N.
Both are dimensionless and hierarchically structured — but they describe different structures. The analogy is *structural*, not *parametric*.

Finding 2: Primes Have No Global Period

The autocorrelation of prime gaps up to 2×10^5 shows no significant periodic structure — the gaps behave approximately like uncorrelated noise, consistent with the Riemann hypothesis.

What primes *do* have:

- **Symmetry structure** (Dirichlet): uniform distribution over all residue classes coprime to m mod m — no accumulation, but symmetry.
- **Characteristic gap distribution**: $\sim$73 % of all prime gaps are multiples of 6, because all primes > 3 lie on $6k \pm 1$.
- **Individual** gcd **structure**: Each prime pair (p, q) has its own gcd$(p-1, q-1)$ that determines $\lambda(N)$.

FFGFT: Three spaces — three different ξ roles

FFGFT distinguishes three mathematically different spaces, in each of which a different ξ parameter acts. These are **not contradictions**, but complementary descriptions of different aspects of the same theory:

Space	Parameter	Domain	Example
Geometric 3D space (torus geometry)	$\xi \approx 1.333 \times 10^{-4}$	Length levels L_N; couplings between spatial scales	Quantum dot scale $N \approx 8$; transistor boundary
Number space / algebra ($\mathbb{Z}/N\mathbb{Z}$, Higgs sector)	$\xi_{\text{Higgs}} \approx 1.038 \times 10^{-5}$	Algebraic field corrections; quantum state space; Bell correlations; decoherence rates	IBM hardware validation; T_2 hierarchy
Number-theoretic space ($\mathbb{Z}/N\mathbb{Z}$ for primes)	*no universal parameter*	Individual gcd$(p-1, q-1)$ structure per prime pair; no global ξ	Shor: period r is N-specific

Consequence for Shor's algorithm: ξ applies to spatial resonances in physical position space — it cannot provide a universal shortcut for period search in $\mathbb{Z}/N\mathbb{Z}$, because this space carries no metric structure that ξ generated. ξ_{Higgs} applies to the algebraic field structure — it appears in the

FFGFT-FFT formulation (Section .2) as a damping factor of the eigenvalue equation, not as a period shortcut.
Structural analogy remains: All three spaces know hierarchies and resonances, all are basis-system independent — but at different levels with different parameters.

Finding 3: Why FFGFT Cannot Directly Form a Better FFT

A global ξ that accelerates period search does not exist — because the primes carry no global resonance parameter.

Why FFT is not directly translatable: FFT requires a complete known data set. Evaluating $f(x) = a^x \bmod N$ for all $x = 0 \ldots 2^n$ explicitly costs $O(N)$ — the advantage would be gone.

The analogue superposition in a physical system (wave field, photonic MZI network) solves this problem — but only if the system evaluates the function $f(x)$ *physically*, not algorithmically.

The core of the problem

The bottleneck in Shor is not the Fourier transformation — it is the **parallel evaluation of** $f(x) = a^x \bmod N$ **for exponentially many** x.
Classical FFT: all $f(x)$ must be known — $O(N)$ effort, unsolvable for RSA.
Quantum computer: prepares $\sum_x |x\rangle|f(x)\rangle$ in $O(n^3)$ steps — through quantum parallelism.
FFGFT eigenvalue problem: does not ask for $f(x)$ for all x, but: at which energies does the potential $V(x) = -\xi \cdot f(x)/N$ have stable resonances? That is a different question — and possibly one that a physical torsion field answers directly, without explicit evaluation.

.4 Worked Example: $N = 15$, $a = 7$ in the FFGFT View

Classical Part: Resonance Strength

Sought: smallest r with $7^r \equiv 1 \pmod{15}$.

The FFGFT resonance strength evaluates each candidate:

$$\text{Res}(r) = \frac{\exp\left(-\xi \cdot r^2/D_{\text{frak}}\right)}{|a^r \bmod N - 1| + 1}. \tag{8}$$

r	$7^r \bmod 15$	Denominator	Resonance strength
1	7	$\lvert 7-1\rvert+1=7$	0.14
2	4	$\lvert 4-1\rvert+1=4$	0.25
3	13	$\lvert 13-1\rvert+1=13$	0.07
4	**1**	$\lvert 1-1\rvert+1=\mathbf{1}$	**1.00**

At $r = 4$ the resonance peaks maximally — the period is found.

Factor Extraction

With $r = 4$ (even):

$$x = 7^{r/2} \bmod 15 = 7^2 \bmod 15 = 49 \bmod 15 = 4, \tag{9}$$

$$\gcd(x-1,\ 15) = \gcd(3,\ 15) = \mathbf{3}, \tag{10}$$

$$\gcd(x+1,\ 15) = \gcd(5,\ 15) = \mathbf{5}. \tag{11}$$

$15 = 3 \times 5$ — correct.

FFGFT Eigenvalue Perspective

In a physical realisation (torsion field on sub-Planck lattice) the system would simultaneously "feel" the entire potential $V(n) = -\xi \cdot (7^n \bmod 15)/15$ — not sequentially for $r = 1, 2, 3, 4$. The resonance at $r = 4$ corresponds to a localised eigenstate of the tight-binding Hamiltonian.

.5 Anderson Localisation: How the Field "Freezes" the Period

Anderson localisation explains the mechanism behind the FFGFT-FFT conjecture.

The pseudorandom potential: $V(n) = a^n \bmod N$ is deterministic but locally jumpy — like a disordered solid-state lattice. Waves propagating through such disorder are scattered at the jumps.

Destructive interference almost everywhere: At most energies E the scattered waves cancel each other — the wave cannot propagate stably.

Constructive interference at $E_k \propto k/r$: Only at resonance frequencies that exactly match the hidden period r of the potential does constructive interference occur — the eigenstate *localises*.

Anderson localisation as period search

The wave function Ψ_n concentrates on a finite region and forms a standing pattern — the period is "frozen" into the system.

The physical system does not compute sequentially $O(N)$ steps — it "locks" into the energetically most stable state that physically represents the solution.

On the sub-Planck scale ($\ell_0 \approx 2 \times 10^{-39}$ m) the lattice is fine enough to interpret the jumps of a^n mod N not as errors but as **geometric information**.

.6 Connection to the Riemann Zeta Function

Pólya-Hilbert Conjecture

The **Pólya-Hilbert conjecture** states: the non-trivial zeros of $\zeta(s)$ are the eigenvalues of a self-adjoint operator $\mathscr{H}$ — the "Riemann Hamiltonian".

No natural physical system has been found so far that generates this spectrum.

FFGFT Candidate on Sub-Planck Lattice

FFGFT proposes: on the sub-Planck scale the fractal spacetime structure ($D_{\text{frak}} = 2.999867$) generates exactly those boundary conditions that the Riemann operator needs.

The potential $V(n) = a^n$ mod N behaves like a chaotic quantum system — and the energy levels of chaotic systems statistically follow the same distributions as the spacings of the zeros of $\zeta(s)$ (Montgomery-Odlyzko law).

Spectral connection: period and primes

If the FFGFT eigenvalue problem and the Riemann zeta function are based on the same physical resonance principle, it follows:
Factorisation is not a combinatorial search problem — it is a **spectroscopic measurement problem**.
One would not have to write a programme, but measure the resonance frequency of a physical spatial region modulated with the potential $V(n)$. The period r — and thus the prime factors — appear as spectral lines.

Status: This connection is mathematically precisely formulable (via the tight-binding Hamiltonian and the Montgomery-Odlyzko law) but not yet proven. It is the open research question of this document — precisely formulated in Section .2.

Why We Cannot Directly Measure This

Thermal noise and the coarse spacetime structure at the Planck scale “smear” the fine resonances. Only below the Planck scale — at $\ell_0 = \xi \cdot \ell_P \approx 2 \times 10^{-39}$ m — is the lattice fine enough to map the jumps of the modulo function without information loss.

At this scale nature performs the QFT “continuously” — as natural dynamics of a wave system that makes periodicities visible through coherent interference.

Overall balance Doc. 176

What is photonically fully realisable:
Every step of Shor’s algorithm is physically executable by a photonic system at room temperature. The FFGFT field equation (2) describes this process mathematically exactly. The MZI network *is* the QFT, not its simulation. The feedforward problem is an open engineering question, not a physical prohibition.

What is not possible on a conventional computer:
Simulating the FFGFT field equation on a digital computer.

The state space problem is exponential — precisely why it needs a physical system that solves the equation by *being* it.

What the test programmes show:
A global ξ parameter that accelerates period search does not exist — the period is determined by gcd$(p-1, q-1)$, not by an external resonance. The FFGFT ξ scan is classical and insufficient for RSA-relevant N.

The open mathematical question:
The FFGFT-FFT as an eigenvalue problem (tight-binding Hamiltonian on sub-Planck lattice) is mathematically precisely formulated but not yet proven — specified in Section .2, connection to the zeta function in Section .6.

Bibliography

[1] Shor, P. W. *Algorithms for quantum computation: discrete logarithms and factoring.* In *Proc. 35th Annual Symp. on Foundations of Computer Science (FOCS)*, 124–134 (IEEE, 1994).

[2] Shor, P. W. *Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer.* SIAM J. Comput. **26**, 1484–1509 (1997).

[3] Griffiths, R. B. & Niu, C.-S. *Semiclassical Fourier transform for quantum computation.* Phys. Rev. Lett. **76**, 3228–3231 (1996).

[4] Knill, E., Laflamme, R. & Milburn, G. J. *A scheme for efficient quantum computation with linear optics.* Nature **409**, 46–52 (2001).

[5] Reck, M. et al. *Experimental realization of any discrete unitary operator.* Phys. Rev. Lett. **73**, 58–61 (1994).

[6] Prevedel, R. et al. *High-speed linear optics quantum computing using active feed-forward.* Nature **445**, 65–69 (2007).

[7] National Institute of Standards and Technology (NIST). *Post-Quantum Cryptography Standards.* FIPS 203–205 (2024). https://csrc.nist.gov/pubs

[8] Fowler, A. G. et al. *Surface codes: towards practical large-scale quantum computation.* Phys. Rev. A **86**, 032324 (2012).

[9] Anderson, P. W. *Absence of diffusion in certain random lattices.* Phys. Rev. **109**, 1492–1505 (1958).

[10] Montgomery, H. L. *The pair correlation of zeros of the zeta function.* Analytic Number Theory, Proc. Sympos. Pure Math. **24**, 181–193 (AMS, 1973).

[11] Berry, M. V. & Keating, J. P. *The Riemann zeros and eigenvalue asymptotics.* SIAM Rev. **41**, 236–266 (1999).

[12] Hofstadter, D. R. *Energy levels and wave functions of Bloch electrons in rational and irrational magnetic fields.* Phys. Rev. B **14**, 2239–2249 (1976).